



ASSIGNMENT PART-2

TECHNOLOGY PARK MALAYSIA

CT042-3-1-IDB

INTRODUCTION TO DATABASES

APU1F2103- APD1F2103-CS-CYB

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1. Introduction

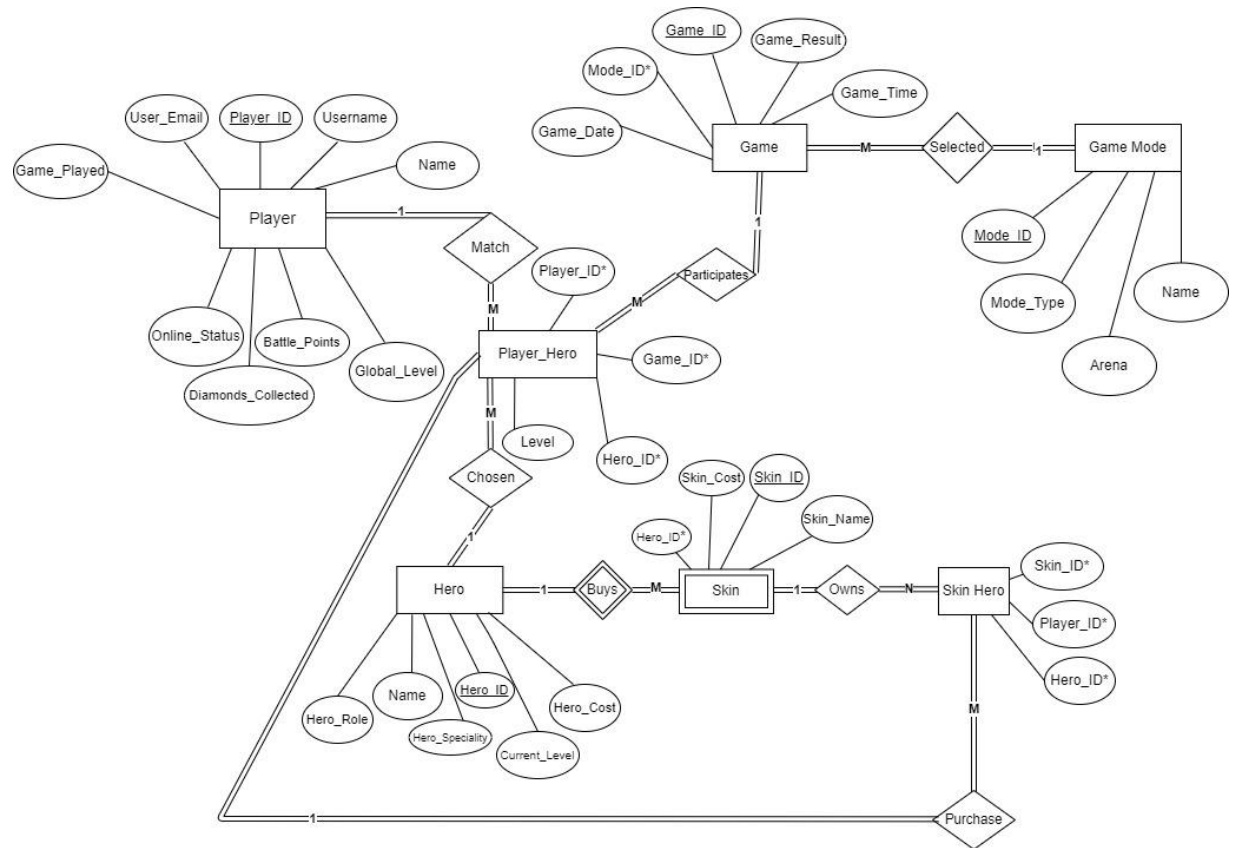
FierceOne Sdn. Bhd., our customer, approached us to help them develop a relational database management system for their new multiplayer online battle arena (MOBA) game, BingBing Legend Mobile Game.

Our group assisted them in resolving their issue by finishing the database design and implementation. This is critical for their new mobile game since they anticipate a large number of players when it launches, thus FierceOne Sdn. Bhd. must have a comprehensive system in place to manage all of the data in an effective and efficient manner.

The database stores all of the game's data, including the player's ID, the Hero they possess, the number of skins each hero has, and much more. This report also contains our finalized Entity Relationship Diagram also known as ERD. An ERD is a sort of structural diagram that is used to map out the database design. It contains various symbols and connections that assist visualize and link information.

We generated a relationship schema based on our finalized Entity-Relationship Diagram, which details the design and structure of the relational database management system we built for BingBing Legend Mobile Game. To eliminate data redundancy, we applied our understanding of SQL Data Definition Language and SQL Data Manipulation Language in the design. In addition, all the relationships in our database were normalized to the third normal form. The database diagram that we created is also provided in our documentation. We made sure to identify the attributes, data types, and constraints utilized in all of our tables in our data dictionary.

2. Final ERD



3. Mapping of ERD to corresponding relational schema

Relational Schema

Player (Player_ID, Player_Name, Username, User_Email, Online_Status, Battle_Points, Diamond_Collected, Game_Played, Global_Level)

Hero (Hero_ID, Hero_Name, Current_Level, Hero_Role, Hero_Speciality, Hero_Cost)

Skin (Skin_ID, Skin_Cost, Skin_Name, Hero_ID(FK))

Game (Game_ID, Game_Date, Game_Time, Game_Result, Mode_ID(FK))

Game_Mode (Mode_ID, Mode_Name, Mode_Type, Arena)

Skin_Hero(Skin_ID(FK1), (Player_ID, Hero_ID) (FK2))

Player_Hero (Player_ID(FK1), Hero_ID(FK2), Game_ID(FK2), Level)

4. Normalization process

Normalization Process

1. **Relation: Player (Player_ID, Player_Name, Username, User_Email, Online_Status, Battle_Points, Diamond_Collected, Game_Played, Global_Level)**

Functional dependencies:

Player_ID → Player_Name, Online_Status, Battle_Points, Diamond_Collected (full dependency)

Player_Name → Username, User_Email, Global_Level, Game_Played (transitive dependency)

3NF

Player2 (Player_ID, Player_Name, Online_Status, Battle_Points, Diamond_Collected)

PlayerInfo (Player_Name, Username, User_Email, Global_Level, Game_Played)

2. **Relation: Hero (Hero_ID, Hero_Name, Current_Level, Hero_Role, Hero_Speciality, Hero_Cost)**

Functional dependencies:

Hero_ID → Hero_Name, Current_Level (full dependency)

Hero_Name → Hero_Role, Hero_Speciality, Hero_Cost (transitive dependency)

3NF

Hero2 (Hero_ID, Hero_Name, Current_Level)

HeroInfo (Hero_Name, Hero_Role, Hero_Speciality, Hero_Cost)

3. **Relation: Skin (Skin_ID, Skin_Name, Skin_Cost, Hero_ID(FK))**

Functional dependencies:

Skin_ID → Skin_Name, Hero_ID (full dependency)

Skin_Name → Skin_Cost (transitive dependency)

3NF

Skin (Skin_ID, Skin_Name, Hero_ID(FK))

SkinInfo (Skin_Name, Skin_Cost)

4. Relation: Game (Game_ID, Game_Date, Game_Time, Game_Result, Mode_ID(FK))

Functional dependencies:

Game_ID $\rightarrow$ Game_Date, Game_Time, Game_Result, Mode_ID (full dependency)

Since there are no partial or transitive dependencies, the Game relation above is already in 3NF.

5. Relation: Game_Mode (Mode_ID, Mode_Name, Mode_Type, Arena)

Functional dependencies:

Mode_ID $\rightarrow$ Mode_Name (full dependency)

Mode_Name $\rightarrow$ Mode_Type, Arena (transitive dependency)

3NF

Game_Mode2 (Mode_ID, Mode_Name)

ModeInfo (Mode_Name, Mode_Type, Arena)

6. Relation: Skin_Hero (Skin_ID(FK1), (Player_ID, Hero_ID) (FK2))

Since there are no non-key attributes in the Skin_Hero relation, it is already in 3NF.

7. Relation: Player_Hero (Player_ID(FK1), Hero_ID(FK2), Game_ID(FK2), Level)

Functional dependencies:

Player_ID, Hero_ID, Game_ID $\rightarrow$ Level (full dependency)

Since there are no partial or transitive dependencies, the Player_Hero relation above is already in 3NF

5. Relational schema in 3NF

Relation in 3NF

Player2 (Player_ID, Player_Name, Online_Status, Battle_Points, Diamond_Collected)

PlayerInfo (Player_Name, Username, User_Email, Global_Level, Game_Played)

Hero2 (Hero_ID, Hero_Name, Current_Level)

HeroInfo (Hero_Name, Hero_Role, Hero_Speciality, Hero_Cost)

Skin (Skin_ID, Skin_Name, Hero_ID(FK))

SkinInfo (Skin_Name, Skin_Cost)

Game (Game_ID, Game_Date, Game_Time, Game_Result, Mode_ID(FK))

Game_Mode2 (Mode_ID, Mode_Name)

ModeInfo (Mode_Name, Mode_Type, Arena)

Skin_Hero (Skin_ID(FK1), (Player_ID, Hero_ID) (FK2))

Player_Hero (Player_ID(FK1), Hero_ID(FK2), Game_ID(FK2), Level)

6. Data dictionary for all tables

DATA DICTIONARY

Table	Attribute	Data Type	Description	Format	Range	Required	PK or FK	Referenced Table
Player2	Player_ID	CHAR (10)	ID of Player	P999	P001-P999	NOT NULL	PK	
	Player_Name	VARCHAR (60)	Name of Player	AAAA		NULL		PlayerInfo
	Online_Status	VARCHAR (60)	Online Status of Player	AAAA		NULL		
	Battle_Points	VARCHAR (60)	Battle Points of Player	9999	0001-9999	NULL		
	Diamond_Collected	INT	Diamond Collected by Player	9999	0001-9999	NULL		
PlayerInfo	Player_Name	VARCHAR (60)	Name of Player	AAAA		NOT NULL	PK	Player2
	Username	VARCHAR (60)	Player's username	AAAA		NULL		
	User_Email	VARCHAR (60)	Player's Email	AAAA		NULL		
	Global_Level	INT	Player Global rank level	9999	0000-9999	NULL		
	Game_Played	INT	Number of Game Played	9999	0000-9999	NULL		
Hero2	Hero_ID	CHAR (10)	Selected hero code	199	101-199	NOT NULL	PK	
	Hero_Name	VARCHAR (60)	Name of hero	AAAAA		NULL		HeroInfo
	Current_Level	INT	Current level of hero2	1	1-99	NULL		
HeroInfo	Hero_Name	VARCHAR (60)	Name of hero	AAAAA		NOT NULL	PK	
	Hero_Role	VARCHAR (60)	Role of the hero	AAAAA		NULL		Hero2

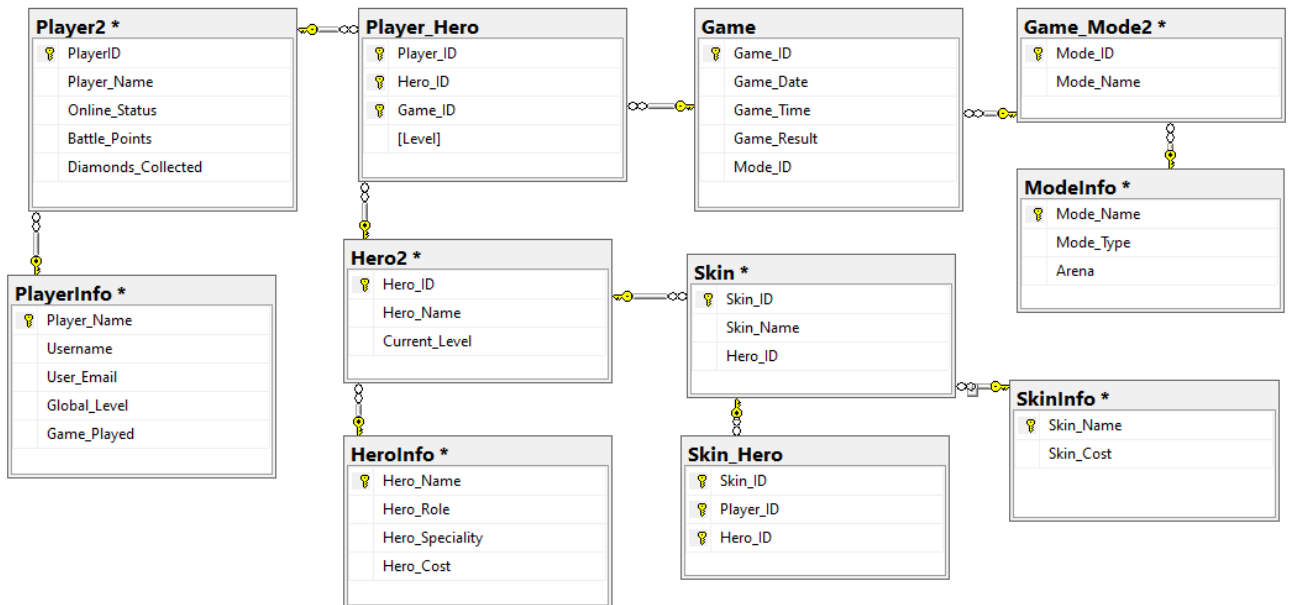
	Hero_Speciality	VARCHAR (60)	Speciality of Hero	AAAAA		NULL		
	Hero_Cost	INT	Cost of Hero	9999	0001-9999	NULL		
Skin	Skin_ID	CHAR (10)	ID of Skin	S99	S01-S99	NOT NULL	PK	Skin_Hero
	Skin_Name	VARCHAR (60)	Name of the Skin	AAAAA		NULL		SkinInfo
	Hero_ID	CHAR (10)	ID of Hero	199	101-199	NULL	FK	
SkinInfo	Skin_Name	VARCHAR (60)	Name of the Skin	AAAAA		NOT NULL	PK	Skin
	Skin_Cost	INT	Cost of Skin	9999	0001-9999	NULL		Skin_Hero

DATA DICTIONARY

Table	Attribute	Data Type	Description	Format	Range	Required	PK or FK	Referenced Table
Game	Game_ID	CHAR (10)	ID of Game	G99999	G00001-G99999	NOT NULL	PK	
	Game_Date	DATE	Date of Game	Yyyy-mm-dd		NULL		Game_Mode2
	Game_Time	TIME	Time of Game	hh:mm:ss		NULL		ModelInfo
	Game_Result	VARCHAR (60)	Result of Game	WIN/LOSE		NULL		
	Mode_ID	CHAR (10)	ID of Mode	GM999	GM001-GM999	NULL	FK	
Game_Mode2	Mode_ID	CHAR (10)	ID of Game_Mode2	GM999	GM001-GM999	NOT NULL	PK	Game
	Mode_Name	VARCHAR (60)	Name of Game_Mode2	AAAAA				ModelInfo
ModelInfo	Mode_Name	VARCHAR (60)	Name of Mode	AAAAA		NOT NULL	PK	
	Mode_Type	VARCHAR (60)	Type of Mode	AAAAA		NULL		Game_Mode2

	Arena	VARCHAR (60)	Type of game mode	AAAAA		NULL		
Skin_Hero	Skin_ID	CHAR (10)	ID of Skin	S99	S01-S99	NOT NULL	PK	
	Player_ID	CHAR (10)	ID of Player	P999	P001-P999	NOT NULL	PK, FK	Skin
	Hero_ID	CHAR (10)	ID of Hero	199	101-199	NOT NULL	PK, FK	SkinInfo
Player_Hero	Player_ID	CHAR (10)	ID of player	P999	P001-P999	NOT NULL	PK, FK	Player2
	Hero_ID	CHAR (10)	ID of Hero	199	101-199	NOT NULL	PK, FK	Hero2
	Game_ID	CHAR (10)	ID of Game	G99999	G00001-G99999	NOT NULL	PK, FK	Game
	Level	INT	Level of Player_Hero	1	1-99	NULL		

7. Database diagram



8. DDL statements + INSERT statements

```
CREATE database BingBingLegends
USE BingBingLegends;
GO

/*Player2*/

CREATE TABLE Player2
(
  PlayerID CHAR(10) PRIMARY KEY NOT NULL,
  Player_Name VARCHAR(60) NULL,
  Online_Status VARCHAR(60) NULL,
  Battle_Points INT NULL,
  Diamonds_Collected INT NULL);

/*PlayerInfo*/

CREATE TABLE PlayerInfo
(
  Player_Name VARCHAR(60) PRIMARY KEY NOT NULL,
  Username VARCHAR(60) NULL,
  User_Email VARCHAR(60) NULL,
  Global_Level INT NULL,
  Game_Played INT NULL);

/*Hero2*/

CREATE TABLE Hero2
(
  Hero_ID CHAR(10) PRIMARY KEY NOT NULL,
  Hero_Name VARCHAR(60) NULL,
  Current_Level INT NULL);

/*HeroInfo*/

CREATE TABLE HeroInfo
(
  Hero_Name VARCHAR(60) PRIMARY KEY NOT NULL,
  Hero_Role VARCHAR(60) NULL,
  Hero_Speciality VARCHAR(60) NULL,
  Hero_Cost INT NULL);
```

```
/*Skin*/
```

```
CREATE TABLE Skin  
  (Skin_ID CHAR(10) PRIMARY KEY NOT NULL,  
   Skin_Name VARCHAR(60) NULL,  
   Hero_ID CHAR(10) NOT NULL REFERENCES Hero2(Hero_ID));
```

```
/*SkinInfo*/
```

```
CREATE TABLE SkinInfo  
  (Skin_Name VARCHAR(60) PRIMARY KEY NOT NULL,  
   Skin_Cost INT NULL);
```

```
/*Game_Mode2*/
```

```
CREATE TABLE Game_Mode2  
  (Mode_ID CHAR(10) PRIMARY KEY NOT NULL,  
   Mode_Name VARCHAR(60) NULL);
```

```
/*Game*/
```

```
CREATE TABLE Game  
  (Game_ID CHAR(10) PRIMARY KEY NOT NULL,  
   Game_Date DATE NULL,  
   Game_Time TIME NULL,  
   Game_Result VARCHAR(60) NULL,  
   Mode_ID CHAR(10) NOT NULL REFERENCES Game_Mode2(Mode_ID));
```

```
/*ModeInfo*/
```

```
CREATE TABLE ModeInfo  
  (Mode_Name VARCHAR(60) PRIMARY KEY NOT NULL,  
   Mode_Type VARCHAR(60) NULL,  
   Arena VARCHAR(60) NULL);
```

```
/*Player_Hero*/
```

```
CREATE TABLE Player_Hero
  (Player_ID CHAR(10) NOT NULL REFERENCES Player2(PlayerID),
   Hero_ID CHAR(10) NOT NULL REFERENCES Hero2(Hero_ID),
   Game_ID CHAR(10) NOT NULL REFERENCES Game(Game_ID),
   Level INT NULL,
   PRIMARY KEY (Player_ID, Hero_ID, Game_ID));
```

```
/*Skin_Hero*/
```

```
CREATE TABLE Skin_Hero
  (Skin_ID CHAR(10) NOT NULL REFERENCES Skin(Skin_ID),
   Player_ID CHAR(10) NOT NULL,
   Hero_ID CHAR(10) NOT NULL,
   PRIMARY KEY (Skin_ID, Player_ID, Hero_ID));
```

```
/*Player2*/
```

```
INSERT INTO Player2 VALUES ('P001', 'Eric White', 'Active', '11', '300');
INSERT INTO Player2 VALUES ('P002', 'Josh Pieters', 'Active', '33', '1500');
INSERT INTO Player2 VALUES ('P003', 'Louis James', 'Inactive', '07', '200');
INSERT INTO Player2 VALUES ('P004', 'Emily Chark', 'Inactive', '15', '1300');
INSERT INTO Player2 VALUES ('P005', 'Rivhard Blair', 'Inactive', '05', '100');
INSERT INTO Player2 VALUES ('P006', 'Chung Garcia', 'Inactive', '19', '1600');
INSERT INTO Player2 VALUES ('P007', 'Daniel Davis', 'Inactive', '25', '500');
INSERT INTO Player2 VALUES ('P008', 'Jack Lee', 'Inactive', '22', '2300');
INSERT INTO Player2 VALUES ('P009', 'Abram Cole', 'Active', '19', '1100');
INSERT INTO Player2 VALUES ('P010', 'Waylon Long', 'Inactive', '12', '1700');
INSERT INTO Player2 VALUES ('P011', 'Chris Wade', 'Inactive', '44', '3300');
INSERT INTO Player2 VALUES ('P012', 'John Alex', 'Active', '39', '3200');
INSERT INTO Player2 VALUES ('P013', 'Raynold Matt', 'Inactive', '22', '1300');
INSERT INTO Player2 VALUES ('P014', 'Eric Cantona', 'Active', '22', '2100');
INSERT INTO Player2 VALUES ('P015', 'Drew Ryan', 'Inactive', '05', '100');
INSERT INTO Player2 VALUES ('P016', 'Reyna Tapia', 'Inactive', '07', '500');
INSERT INTO Player2 VALUES ('P017', 'Zane Gomez', 'Inactive', '21', '1900');
INSERT INTO Player2 VALUES ('P018', 'Bobbi Pope', 'Active', '25', '1350');
INSERT INTO Player2 VALUES ('P019', 'Allison Baker', 'Inactive', '21', '900');
INSERT INTO Player2 VALUES ('P020', 'Ashley Young', 'Inactive', '16', '500');
```

```
/*PlayerInfo*/
```

```
INSERT INTO PlayerInfo VALUES ('Eric White','ericpieters','eric123@gmail.com','23','11');
INSERT INTO PlayerInfo VALUES ('Josh Pieters','joshpietss','jpieters01@gmail.com','50','71');
INSERT INTO PlayerInfo VALUES ('Louis James','JLouis','jameslouis@gmail.com','02','01');
INSERT INTO PlayerInfo VALUES ('Emily Chark','Emilyyyy','clarkemy@gmail.com','31','47');
INSERT INTO PlayerInfo VALUES ('Rivhard Blair','BlairR','blair001@gmail.com','33','50');
INSERT INTO PlayerInfo VALUES ('Chung Garcia','ChungGarcia','garciachung@gmail.com','53','83');
INSERT INTO PlayerInfo VALUES ('Daniel Davis','DDaniel','ddaniel10@gmail.com','62','70');
INSERT INTO PlayerInfo VALUES ('Jack Lee','LeeJ','leejack@gmail.com','55','66');
INSERT INTO PlayerInfo VALUES ('Abram Cole','ColeRam','abramcole@gmail.com','33','41');
INSERT INTO PlayerInfo VALUES ('Waylon Long','Longlon','longlon@gmail.com','23','20');
INSERT INTO PlayerInfo VALUES ('Chris Wade','Chrissade','chriswade110@gmail.com','51','82');
INSERT INTO PlayerInfo VALUES ('John Alex','jonnnylex','johnalex01@gmail.com','73','113');
INSERT INTO PlayerInfo VALUES ('Raynold Matt','mattynold','mattraynold22@gmail.com','41','60');
INSERT INTO PlayerInfo VALUES ('Eric Cantona','Cantonnaric','cantonaeric@gmail.com','47','55');
INSERT INTO PlayerInfo VALUES ('Drew Ryan','RDrew','rdrew0001@gmail.com','03','02');
INSERT INTO PlayerInfo VALUES ('Reyna Tapia','Tapiana','tapianaaaaa@gmail.com','03','04');
INSERT INTO PlayerInfo VALUES ('Zane Gomez','ZGomez','zgomezzy1@gmail.com','41','72');
INSERT INTO PlayerInfo VALUES ('Bobbi Pope','bobby','bobbypope0123@gmail.com','45','60');
INSERT INTO PlayerInfo VALUES ('Allison Baker','bakkeryson','allison321@gmail.com','73','109');
INSERT INTO PlayerInfo VALUES ('Ashley Young','ashyoung','youngashley30@gmail.com','04','03');
```

```
--Hero2 table
```

```
INSERT INTO Hero2 VALUES (101,'Dante',2);
INSERT INTO Hero2 VALUES (102,'Hestia',8);
INSERT INTO Hero2 VALUES (103,'Ophelia',4);
INSERT INTO Hero2 VALUES (104,'Sterben',1);
INSERT INTO Hero2 VALUES (105,'Gaia',3);
INSERT INTO Hero2 VALUES (106,'Uranus',2);
INSERT INTO Hero2 VALUES (107,'Sierra',4);
INSERT INTO Hero2 VALUES (108,'Medusa',1);
INSERT INTO Hero2 VALUES (109,'Ymer',2);
INSERT INTO Hero2 VALUES (110,'Asgard',3);
INSERT INTO Hero2 VALUES (111,'Hyperion',1);
INSERT INTO Hero2 VALUES (112,'Typhon',2);
INSERT INTO Hero2 VALUES (113,'Lyssa',5);
INSERT INTO Hero2 VALUES (114,'Eudora',7);
INSERT INTO Hero2 VALUES (115,'Aura',5);
```



```
--HeroInfo
```

```
INSERT INTO HeroInfo VALUES ('Dante','Fighter','Charge',700);
INSERT INTO HeroInfo VALUES ('Hestia','Mage','Efficient Spells',850);
INSERT INTO HeroInfo VALUES ('Ophelia','Mage','Wisdom',900);
INSERT INTO HeroInfo VALUES ('Sterben','Assassin','Dodging',1000);
INSERT INTO HeroInfo VALUES ('Gaia','Tank','Fortress',1200);
INSERT INTO HeroInfo VALUES ('Uranus','Fighter','Knockback',1450);
INSERT INTO HeroInfo VALUES ('Sierra','Archer','Triple Shot',1150);
INSERT INTO HeroInfo VALUES ('Medusa','Mage','Rapid Fire',1750);
INSERT INTO HeroInfo VALUES ('Ymer','Assassin','Invicibility',1500);
INSERT INTO HeroInfo VALUES ('Asgard','Fighter','Life Steal',1900);
INSERT INTO HeroInfo VALUES ('Hyperion','Tank','Reflection',2300);
INSERT INTO HeroInfo VALUES ('Typhon','Archer','Bouncy Arrow',2200);
INSERT INTO HeroInfo VALUES ('Lyssa','Archer','Target Tracer',3000);
INSERT INTO HeroInfo VALUES ('Eudora','Fighter','Berserker',2250);
INSERT INTO HeroInfo VALUES ('Aura','Healer','Revive',3500);
```

```
/*Game Mode2*/
```

```
INSERT INTO Game_Mode2 VALUES ('GM001','INFINITE MELEE');
INSERT INTO Game_Mode2 VALUES ('GM002','MAN-MACHINE BATTLE');
INSERT INTO Game_Mode2 VALUES ('GM003','TEAM MATCH');
```

```
/*Game*/
```

```
INSERT INTO Game VALUES ('G00001', '2021-01-22', '12:23', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00002', '2021-01-14', '17:01', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00003', '2021-04-06', '21:30', 'LOSE', 'GM001');
INSERT INTO Game VALUES ('G00004', '2021-08-12', '14:46', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00005', '2021-09-25', '06:28', 'LOSE', 'GM003');
INSERT INTO Game VALUES ('G00006', '2021-03-15', '09:59', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00007', '2021-02-15', '13:12', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00008', '2021-07-23', '17:56', 'LOSE', 'GM001');
INSERT INTO Game VALUES ('G00009', '2021-06-26', '11:26', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00010', '2021-02-28', '02:18', 'LOSE', 'GM001');
INSERT INTO Game VALUES ('G00011', '2021-03-15', '23:35', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00012', '2021-08-15', '15:16', 'LOSE', 'GM001');
INSERT INTO Game VALUES ('G00013', '2021-04-17', '19:35', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00014', '2021-10-10', '10:17', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00015', '2021-12-29', '23:45', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00016', '2021-11-24', '00:00', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00017', '2021-08-31', '16:12', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00018', '2021-03-15', '19:04', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00019', '2021-02-04', '12:28', 'LOSE', 'GM003');
INSERT INTO Game VALUES ('G00020', '2021-03-15', '01:23', 'WIN', 'GM001');
INSERT INTO Game VALUES ('G00021', '2021-06-21', '17:46', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00022', '2021-03-08', '02:50', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00023', '2021-06-21', '11:17', 'LOSE', 'GM003');
INSERT INTO Game VALUES ('G00024', '2021-04-01', '20:10', 'WIN', 'GM001');
INSERT INTO Game VALUES ('G00025', '2021-08-05', '14:12', 'WIN', 'GM002');
INSERT INTO Game VALUES ('G00026', '2021-09-07', '11:01', 'WIN', 'GM001');
INSERT INTO Game VALUES ('G00027', '2021-10-28', '10:31', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00028', '2021-04-18', '12:06', 'WIN', 'GM003');
INSERT INTO Game VALUES ('G00029', '2021-11-11', '23:37', 'LOSE', 'GM003');
INSERT INTO Game VALUES ('G00030', '2021-12-25', '20:43', 'WIN', 'GM001');
```

```
/*ModeInfo*/
```

```
INSERT INTO ModeInfo VALUES ('INFINITE MELEE', 'BRAWL MODE', 'ABYSS OF DREAMLAD');
INSERT INTO ModeInfo VALUES ('MAN-MACHINE BATTLE', 'HUMAN VS AI MODE', 'CLONE FIGHT');
INSERT INTO ModeInfo VALUES ('TEAM MATCH', 'CLASSIC MODE', 'FIGHT FOR FIRE');
```

```
/*Player_Hero*/
```

```
INSERT INTO Player_Hero VALUES ('P001','101','G00001','5');
INSERT INTO Player_Hero VALUES ('P002','102','G00001','18');
INSERT INTO Player_Hero VALUES ('P003','103','G00001','4');
INSERT INTO Player_Hero VALUES ('P004','104','G00001','8');
INSERT INTO Player_Hero VALUES ('P005','105','G00001','5');
INSERT INTO Player_Hero VALUES ('P006','106','G00002','20');
INSERT INTO Player_Hero VALUES ('P007','107','G00002','12');
INSERT INTO Player_Hero VALUES ('P008','102','G00002','1');
INSERT INTO Player_Hero VALUES ('P009','109','G00002','2');
INSERT INTO Player_Hero VALUES ('P010','110','G00002','10');
INSERT INTO Player_Hero VALUES ('P011','111','G00003','1');
INSERT INTO Player_Hero VALUES ('P012','102','G00003','2');
INSERT INTO Player_Hero VALUES ('P013','113','G00003','5');
INSERT INTO Player_Hero VALUES ('P014','114','G00003','7');
INSERT INTO Player_Hero VALUES ('P015','115','G00003','5');
```

```
/*Skin*/
```

```
INSERT INTO Skin VALUES ('S01','Assasin Darius','101');
INSERT INTO Skin VALUES ('S02','Battlecast Elise','102');
INSERT INTO Skin VALUES ('S03','Blacksmith Mordekaiser','103');
INSERT INTO Skin VALUES ('S04','Full Metal Veigar','104');
INSERT INTO Skin VALUES ('S05','Wonderland Riven','105');
INSERT INTO Skin VALUES ('S06','Vandal Yorick','106');
INSERT INTO Skin VALUES ('S07','Debonair Urgot','107');
INSERT INTO Skin VALUES ('S08','Barbarian Karthus','108');
INSERT INTO Skin VALUES ('S09','Runeguard Brand','109');
INSERT INTO Skin VALUES ('S10','Divine Evelyn','110');
INSERT INTO Skin VALUES ('S11','Spooky Ziggs','111');
INSERT INTO Skin VALUES ('S12','Explorer Xerath','112');
INSERT INTO Skin VALUES ('S13','Tango Galio','113');
INSERT INTO Skin VALUES ('S14','Forsaken Ryze','114');
INSERT INTO Skin VALUES ('S15','Franken Zyra','115');
```

```
/*Skin_Hero*/
```

```
INSERT INTO Skin_Hero VALUES ('S01','P001','101');
INSERT INTO Skin_Hero VALUES ('S02','P002','102');
INSERT INTO Skin_Hero VALUES ('S03','P003','103');
INSERT INTO Skin_Hero VALUES ('S04','P004','104');
INSERT INTO Skin_Hero VALUES ('S05','P005','105');
INSERT INTO Skin_Hero VALUES ('S06','P006','106');
INSERT INTO Skin_Hero VALUES ('S07','P007','107');
INSERT INTO Skin_Hero VALUES ('S08','P008','108');
INSERT INTO Skin_Hero VALUES ('S09','P009','109');
INSERT INTO Skin_Hero VALUES ('S10','P010','110');
INSERT INTO Skin_Hero VALUES ('S11','P011','111');
INSERT INTO Skin_Hero VALUES ('S12','P012','112');
INSERT INTO Skin_Hero VALUES ('S13','P013','113');
INSERT INTO Skin_Hero VALUES ('S14','P014','114');
INSERT INTO Skin_Hero VALUES ('S15','P015','115');
```

```
/*SkinInfo*/
```

```
INSERT INTO SkinInfo VALUES ('Assasin Darius','500');
INSERT INTO SkinInfo VALUES ('Battlecast Elise','600');
INSERT INTO SkinInfo VALUES ('Blacksmith Mordekaiser','700');
INSERT INTO SkinInfo VALUES ('Full Metal Veigar','800');
INSERT INTO SkinInfo VALUES ('Wonderland Riven','900');
INSERT INTO SkinInfo VALUES ('Vandal Yorick','1000');
INSERT INTO SkinInfo VALUES ('Debonair Urgot','1100');
INSERT INTO SkinInfo VALUES ('Barbarian Karthus','1200');
INSERT INTO SkinInfo VALUES ('Runeguard Brand','1300');
INSERT INTO SkinInfo VALUES ('Divine Evelynn','1400');
INSERT INTO SkinInfo VALUES ('Spooky Ziggs','1500');
INSERT INTO SkinInfo VALUES ('Explorer Xerath','1600');
INSERT INTO SkinInfo VALUES ('Tango Galio','1700');
INSERT INTO SkinInfo VALUES ('Forsaken Ryze','1800');
INSERT INTO SkinInfo VALUES ('Franken Zyra','1900');
```

9. Screen capture of data in all tables

Player2

	PlayerID	Player_Na...	Online_Stat...	Battle_Points	Diamonds_...
▶	P001	Eric White	Active	11	300
	P002	Josh Pieters	Active	33	1500
	P003	Louis James	Inactive	7	200
	P004	Emily Chark	Inctive	15	1300
	P005	Rivhard Blair	Inactive	5	100
	P006	Chung Garcia	Inctive	19	1600
	P007	Daniel Davis	Inactive	25	500
	P008	Jack Lee	Inactive	22	2300
	P009	Abram Cole	Active	19	1100
	P010	Waylon Long	Inactive	12	1700
	P011	Chris Wade	Inactive	44	3300
	P012	John Alex	Active	39	3200
	P013	Raynold Matt	Inactive	22	1300
	P014	Eric Cantona	Active	22	2100
	P015	Drew Ryan	Inactive	5	100
	P016	Reyna Tapia	Inactive	7	500
	P017	Zane Gomez	Inactive	21	1900
	P018	Bobbi Pope	Active	25	1350
	P019	Allison Baker	Inactive	21	900
	P020	Ashley Young	Inactive	16	500
*	NULL	NULL	NULL	NULL	NULL

PlayerInfo

	Player_Name	Username	User_Email	Global_Level	Game_Play...
►	Abram Cole	ColeRam	abramcole@gmail.com	33	41
	Allison Baker	bakkeryson	allison321@gmail.com	73	109
	Ashley Young	ashyoung	youngashley30@gmail.com	4	3
	Bobbi Pope	bobby	bobbypope0123@gmail.com	45	60
	Chris Wade	Chrissade	chriswade110@gmail.com	51	82
	Chung Garcia	ChungGarcia	garciachung@gmail.com	53	83
	Daniel Davis	DDaniel	ddaniel10@gmail.com	62	70
	Drew Ryan	RDrew	rdrew0001@gmail.com	3	2
	Emily Chark	Emillyyy	clarkemy@gmail.com	31	47
	Eric Cantona	Cantonnaric	cantonaeric@gmail.com	47	55
	Eric White	ericpieters	eric123@gmail.com	23	11
	Jack Lee	LeeJ	leejack@gmail.com	55	66
	John Alex	jonnnylex	johnalex01@gmail.com	73	113
	Josh Pieters	joshpietss	jpieters01@gmail.com	50	71
	Louis James	JLouis	jameslouis@gmail.com	2	1
	Raynold Matt	mattynold	matraynold22@gmail.com	41	60
	Reyna Tapia	Tapiana	tapianaaaaa@gmail.com	3	4
	Rivhard Blair	BlairR	blair001@gmail.com	33	50
	Waylon Long	Longlon	longlon@gmail.com	23	20
	Zane Gomez	ZGomez	zgomezyy1@gmail.com	41	72
*	NULL	NULL	NULL	NULL	NULL

Hero2

	Hero_ID	Hero_Name	Current_Lev...
►	101	Dante	2
	102	Hestia	8
	103	Ophelia	4
	104	Sterben	1
	105	Gaia	3
	106	Uranus	2
	107	Sierra	4
	108	Medusa	1
	109	Ymer	2
	110	Asgard	3
	111	Hyperion	1
	112	Typhon	2
	113	Lyssa	5
	114	Eudora	7
	115	Aura	5
*	NULL	NULL	NULL

HeroInfo

	Hero_Name	Hero_Role	Hero_Speci...	Hero_Cost
►	Asgard	Fighter	Life Steal	1900
	Aura	Healer	Revive	3500
	Dante	Fighter	Charge	700
	Eudora	Fighter	Berserker	2250
	Gaia	Tank	Fortress	1200
	Hestia	Mage	Efficient Sp...	850
	Hyperion	Tank	Reflection	2300
	Lyssa	Archer	Target Tracer	3000
	Medusa	Mage	Rapid Fire	1750
	Ophelia	Mage	Wisdom	900
	Sierra	Archer	Triple Shot	1150
	Sterben	Assassin	Dodging	1000
	Typhon	Archer	Bouncy Arro...	2200
	Uranus	Fighter	Knockback	1450
	Ymer	Assassin	Invicibility	1500
*	NULL	NULL	NULL	NULL

Skin

	Skin_ID	Skin_Name	Hero_ID
►	S01	Assasin Darius	101
	S02	Battlecast Elise	102
	S03	Blacksmith Mordekaiser	103
	S04	Full Metal Veigar	104
	S05	Wonderland Riven	105
	S06	Vandal Yorick	106
	S07	Debonair Urgot	107
	S08	Barbarian Karthus	108
	S09	Runeguard Brand	109
	S10	Divine Evelynn	110
	S11	Spooky Ziggs	111
	S12	Explorer Xerath	112
	S13	Tango Galio	113
	S14	Forsaken Ryze	114
	S15	Franken Zyra	115
*	NULL	NULL	NULL

SkinInfo

	Skin_Name	Skin_Cost
►	Assasin Darius	500
	Barbarian Karthus	1200
	Battlecast Elise	600
	Blacksmith Mordekaiser	700
	Debonair Urgot	1100
	Divine Evelynn	1400
	Explorer Xerath	1600
	Forsaken Ryze	1800
	Franken Zyra	1900
	Full Metal Veigar	800
	Runeguard Brand	1300
	Spooky Ziggs	1500
	Tango Galio	1700
	Vandal Yorick	1000
	Wonderland Riven	900
*	NULL	NULL

Game_Mode2

	Mode_ID	Mode_Name
▶	GM001	INFINITE MELEE
	GM002	MAN-MACHINE BATTLE
	GM003	TEAM MATCH
*	NULL	NULL

ModelInfo

	Mode_Name	Mode_Type	Arena
▶	INFINITE MELEE	BRAWL MODE	ABYSS OF DREAMLAD
	MAN-MACHINE BATTLE	HUMAM VS AI MODE	CLONE FIGHT
	TEAM MATCH	CLASSIC MODE	FIGHT FOR FIRE
*	NULL	NULL	NULL

Player_Hero

	Player_ID	Hero_ID	Game_ID	Level
▶	P001	101	G00001	5
	P002	102	G00001	18
	P003	103	G00001	4
	P004	104	G00001	8
	P005	105	G00001	5
	P006	106	G00002	20
	P007	107	G00002	12
	P008	102	G00002	1
	P009	109	G00002	2
	P010	110	G00002	10
	P011	111	G00003	1
	P012	102	G00003	2
	P013	113	G00003	5
	P014	114	G00003	7
	P015	115	G00003	5
*	NULL	NULL	NULL	NULL

Game

	Game_ID	Game_Date	Game_Time	Game_Result	Mode_ID
▶	G00001	2021-01-22	12:23:00	WIN	GM003
	G00002	2021-01-14	17:01:00	WIN	GM002
	G00003	2021-04-06	21:30:00	LOSE	GM001
	G00004	2021-08-12	14:46:00	WIN	GM003
	G00005	2021-09-25	06:28:00	LOSE	GM003
	G00006	2021-03-15	09:59:00	WIN	GM002
	G00007	2021-02-15	13:12:00	WIN	GM002
	G00008	2021-07-23	17:56:00	LOSE	GM001
	G00009	2021-06-26	11:26:00	WIN	GM002
	G00010	2021-02-28	02:18:00	LOSE	GM001
	G00011	2021-03-15	23:35:00	WIN	GM003
	G00012	2021-08-15	15:16:00	LOSE	GM001
	G00013	2021-04-17	19:35:00	WIN	GM003
	G00014	2021-10-10	10:17:00	WIN	GM002
	G00015	2021-12-29	23:45:00	WIN	GM003
	G00016	2021-11-24	00:00:00	WIN	GM002
	G00017	2021-08-31	16:12:00	WIN	GM002
	G00018	2021-03-15	19:04:00	WIN	GM003
	G00019	2021-02-04	12:28:00	LOSE	GM003
	G00020	2021-03-15	01:23:00	WIN	GM001
	G00021	2021-06-21	17:46:00	WIN	GM003
	G00022	2021-03-08	02:50:00	WIN	GM002
	G00023	2021-06-21	11:17:00	LOSE	GM003
	G00024	2021-04-01	20:10:00	WIN	GM001
	G00025	2021-08-05	14:12:00	WIN	GM002
	G00026	2021-09-07	11:01:00	WIN	GM001
	G00027	2021-10-28	10:31:00	WIN	GM003
	G00028	2021-04-18	12:06:00	WIN	GM003
	G00029	2021-11-11	23:37:00	LOSE	GM003
	G00030	2021-12-25	20:43:00	WIN	GM001
*	NULL	NULL	NULL	NULL	NULL

Skin_Hero

	Skin_ID	Player_ID	Hero_ID
►	S01	P001	101
	S02	P002	102
	S03	P003	103
	S04	P004	104
	S05	P005	105
	S06	P006	106
	S07	P007	107
	S08	P008	108
	S09	P009	109
	S10	P010	110
	S11	P011	111
	S12	P012	112
	S13	P013	113
	S14	P014	114
	S15	P015	115
*	NULL	NULL	NULL

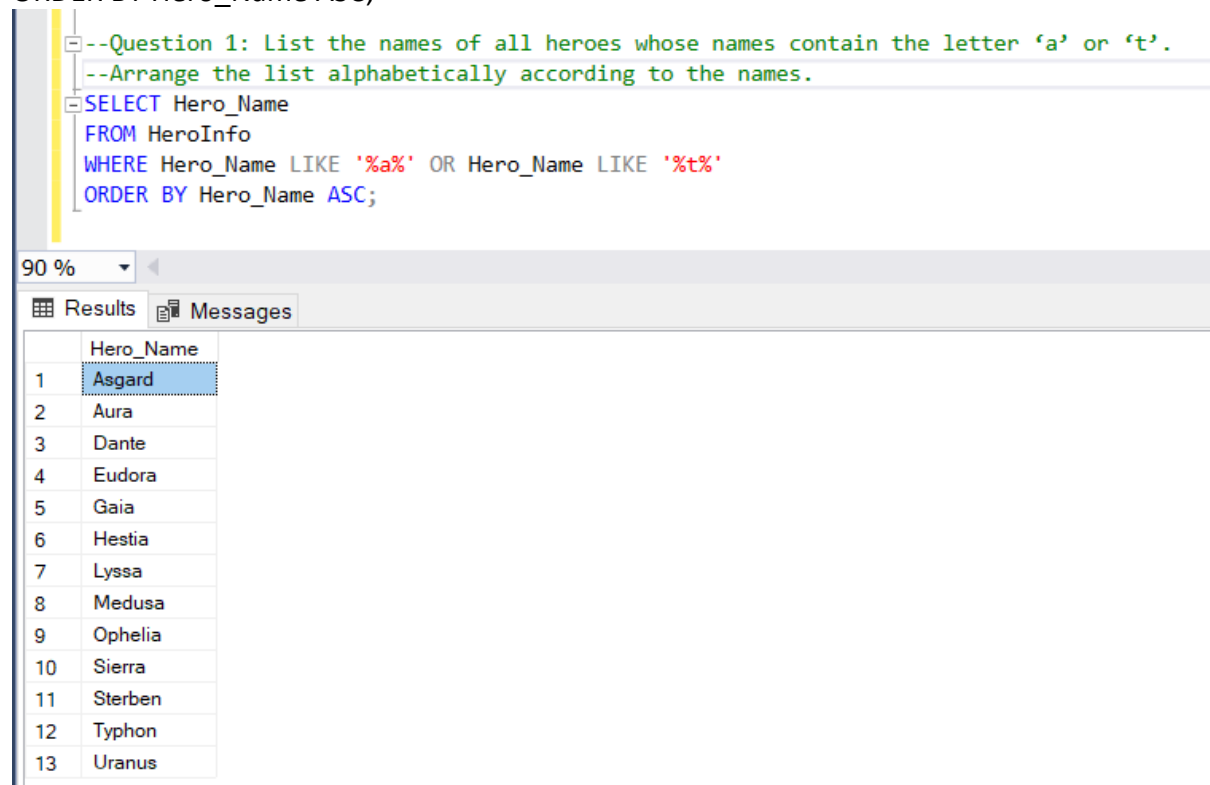
10. SQL statements (DML)

Student 1: (Abdullah Al Mukit TP063405)

Question 1: List the names of all heroes whose names contain the letter 'a' or 't'. Arrange the list alphabetically according to the names.

Solution:

```
SELECT Hero_Name
FROM HeroInfo
WHERE Hero_Name LIKE '%a%' OR Hero_Name LIKE '%t%'
ORDER BY Hero_Name ASC;
```



```
--Question 1: List the names of all heroes whose names contain the letter 'a' or 't'.
--Arrange the list alphabetically according to the names.
SELECT Hero_Name
FROM HeroInfo
WHERE Hero_Name LIKE '%a%' OR Hero_Name LIKE '%t%'
ORDER BY Hero_Name ASC;
```

	Hero_Name
1	Asgard
2	Aura
3	Dante
4	Eudora
5	Gaia
6	Hestia
7	Lyssa
8	Medusa
9	Ophelia
10	Sierra
11	Sterben
12	Typhon
13	Uranus

Question 2: List the details of players who own any heroes with “Charge” specialty

Solution:

```
SELECT HeroInfo.Hero_Name, HeroInfo.Hero_Speciality, Hero2.Hero_ID,
Player_Hero.Player_ID, Player2.Player_Name
FROM HeroInfo
JOIN Hero2 ON HeroInfo.Hero_Name = Hero2.Hero_Name
JOIN Player_Hero ON Hero2.Hero_ID = Player_Hero.Hero_ID
Join Player2 ON Player_Hero.Player_ID = Player2.PlayerID
WHERE HeroInfo.Hero_Speciality = 'Charge';
```

```
--Question 2: List the details of players who own any heroes with "Charge" specialty
SELECT HeroInfo.Hero_Name, HeroInfo.Hero_Speciality, Hero2.Hero_ID, Player_Hero.Player_ID, Player2.Player_Name
FROM HeroInfo
JOIN Hero2 ON HeroInfo.Hero_Name = Hero2.Hero_Name
JOIN Player_Hero ON Hero2.Hero_ID = Player_Hero.Hero_ID
Join Player2 ON Player_Hero.Player_ID = Player2.PlayerID
WHERE HeroInfo.Hero_Speciality = 'Charge';
```

90 %

Results Messages

	Hero_Name	Hero_Speciality	Hero_ID	Player_ID	Player_Name
1	Dante	Charge	101	P001	Eric White

Question 3: List the ids, names and levels of all heroes owned by the player named "John Alex".

Solution:

```
SELECT Player_Hero.Hero_ID, Hero2.Hero_Name, Player_Hero.Level
FROM Player2 JOIN Player_Hero
ON Player2.PlayerID = Player_Hero.Player_ID
JOIN Hero2 ON Player_Hero.Hero_ID = Hero2.Hero_ID
WHERE Player2.Player_Name = 'John Alex';
```

```
--Question 3: List the ids, names and levels of all heroes owned by the player named "John Alex".
SELECT Player_Hero.Hero_ID, Hero2.Hero_Name, Player_Hero.Level
FROM Player2 JOIN Player_Hero
ON Player2.PlayerID = Player_Hero.Player_ID
JOIN Hero2 ON Player_Hero.Hero_ID = Hero2.Hero_ID
WHERE Player2.Player_Name = 'John Alex';
```

90 %

Results Messages

	Hero_ID	Hero_Name	Level
1	102	Hestia	2

Student 2: (Saw Chiou Ming TP063734)

Question1: List the game ids, and game times, game dates, for all games played between '01/03/2021' and '31/03/2021' in descending order of game dates.

Solution:

```
SELECT Game_ID, Game_Time, Game_Date
FROM Game
WHERE Game_Date BETWEEN '2021-03-01' AND '2021-03-31'
ORDER BY Game_Date DESC;
```

--Question1: List the game ids, and game times, game dates, for all games played between '01/03/2021' and '31/03/2021' in descending order of game dates.

```
SELECT Game_ID, Game_Time, Game_Date
FROM Game
WHERE Game_Date BETWEEN '2021-03-01' AND '2021-03-31'
ORDER BY Game_Date DESC;
```

	Game_ID	Game_Time	Game_Date
1	G00014	10:17:00.0000000	2021-03-28
2	G00018	19:04:00.0000000	2021-03-15
3	G00020	01:23:00.0000000	2021-03-15
4	G00006	09:59:00.0000000	2021-03-15
5	G00011	23:35:00.0000000	2021-03-15
6	G00022	02:50:00.0000000	2021-03-08

Question 2: List the details of players who own any heroes whose levels are more than 3.

Solution:

```
SELECT P.playerID , PI.*
FROM Player2 P JOIN PlayerInfo PI ON P.Player_Name = PI.Player_Name
JOIN Player_Hero PH ON PH.Player_ID = P.PlayerID
JOIN Hero2 H ON H.Hero_ID = PH.Hero_ID
WHERE H.Current_Level > 3;
```

--Question 2: List the details of players who own any heroes whose levels are more than 3.

```
SELECT P.playerID , PI.*
FROM Player2 P JOIN PlayerInfo PI ON P.Player_Name = PI.Player_Name
JOIN Player_Hero PH ON PH.Player_ID = P.PlayerID
JOIN Hero2 H ON H.Hero_ID = PH.Hero_ID
WHERE H.Current_Level > 3;
```

	playerID	Player_Name	Username	User_Email	Global_Level	Game_Played
1	P002	Josh Pieters	joshpietss	jpieters01@gmail.com	50	71
2	P003	Louis James	JLouis	jameslouis@gmail.com	2	1
3	P007	Daniel Davis	DDaniel	ddaniel10@gmail.com	62	70
4	P008	Jack Lee	LeeJ	leejack@gmail.com	55	66
5	P012	John Alex	jonnnylex	johnalex01@gmail.com	73	113
6	P013	Raynold Matt	mattynold	mattraynold22@gmail.com	41	60
7	P014	Eric Cantona	Cannonnaric	cantonaeric@gmail.com	47	55
8	P015	Drew Ryan	RDrew	rdrew0001@gmail.com	3	2

Question 3: List the ids and names of all heroes who has any skin priced (in battle points) higher than the average price of all skins, and show by how much.

Solution:

```
SELECT H.Hero_ID, H.Hero_Name, SI.Skin_Cost
FROM skinInfo SI JOIN skin S ON SI.Skin_Name = S.Skin_Name
JOIN skin_hero SH ON S.Skin_ID = SH.Skin_ID
JOIN hero2 H ON SH.Hero_ID = H.Hero_ID
WHERE SI.Skin_Cost > (SELECT AVG(Skin_Cost) FROM SkinInfo);
```

```
--Question 3: List the ids and names of all heroes who has any skin priced (in battle points) higher than the average price of all skins, and show by how much.
SELECT H.Hero_ID, H.Hero_Name, SI.Skin_Cost
FROM skinInfo SI JOIN skin S ON SI.Skin_Name = S.Skin_Name
JOIN skin_hero SH ON S.Skin_ID = SH.Skin_ID
JOIN hero2 H ON SH.Hero_ID = H.Hero_ID
WHERE SI.Skin_Cost > (SELECT AVG(Skin_Cost) FROM SkinInfo);
|
```

90 %

Results Messages

	Hero_ID	Hero_Name	Skin_Cost
1	109	Ymer	1300
2	110	Asgard	1400
3	111	Hyperion	1500
4	112	Typhon	1600
5	113	Lyssa	1700
6	114	Eudora	1800
7	115	Aura	1900

Student 3: SM Rabeya Amin Ikra (TP058834)

Question1: List the names and email addresses of all the players who have played at least 5 games.

Solution:

```
SELECT Player_Name,User_Email,Game_Played
FROM PlayerInfo
WHERE Game_Played >=5
ORDER BY Player_Name ASC;
```

--Question1: List the names and email addresses of all the players who have played atleast 5 games.

```
SELECT Player_Name,User_Email,Game_Played
FROM PlayerInfo
WHERE Game_Played >=5
ORDER BY Player_Name ASC;
```

90 %

Results Messages

	Player_Name	User_Email	Game_Played
1	Abram Cole	abramcole@gmail.com	41
2	Allison Baker	allison321@gmail.com	109
3	Bobbi Pope	bobbypope0123@gmail.com	60
4	Chris Wade	chrismwade110@gmail.com	82
5	Chung Garcia	garciachung@gmail.com	83
6	Daniel Davis	ddaniel10@gmail.com	70
7	Emily Chark	clarkemy@gmail.com	47
8	Eric Cantona	cantonaeric@gmail.com	55
9	Eric White	eric123@gmail.com	11
10	Jack Lee	leejack@gmail.com	66
11	John Alex	johnalex01@gmail.com	113
12	Josh Pieters	jpieters01@gmail.com	71
13	Raynold Matt	mattraynold22@gmail.com	60
14	Rivhard Blair	blair001@gmail.com	50
15	Waylon Long	longlon@gmail.com	20
16	Zane Gomez	zgomezzy1@gmail.com	72

Question 2: List the hero ID, Hero name and Total Number of skins belonging to each hero.

Solution:

```
SELECT H.Hero_ID,H.Hero_Name,COUNT(S.Skin_ID) AS 'TOTAL NUMBER OF
SKINS'
FROM Hero2 H inner join Skin S ON H.Hero_ID = S.Hero_ID
GROUP BY H.Hero_ID, H.Hero_Name
```



```
--Question 2: List the hero ID, Hero name and Total Number of skins belonging to each hero.
SELECT H.Hero_ID,H.Hero_Name,COUNT(S.Skin_ID) AS 'TOTAL NUMBER OF SKINS'
FROM Hero2 H inner join Skin S ON H.Hero_ID = S.Hero_ID
GROUP BY H.Hero_ID, H.Hero_Name;
```

90 %

Results Messages

	Hero_ID	Hero_Name	TOTAL NUMBER OF SKINS
1	101	Dante	1
2	102	Hestia	1
3	103	Ophelia	1
4	104	Sterben	1
5	105	Gaia	1
6	106	Uranus	1
7	107	Sierra	1
8	108	Medusa	1
9	109	Ymer	1
10	110	Asgard	1
11	111	Hyperion	1
12	112	Typhon	1
13	113	Lyssa	1
14	114	Eudora	1
15	115	Aura	1

Question 3: List the details of all players who did not play any games.

Solution:

```
SELECT T.*,P.Global_Level,P.Game_Played,P.User_Email,P.Username
FROM PlayerInfo P INNER JOIN Player2 T ON P.Player_Name = T.Player_Name
WHERE Game_Played =0
```

```
--Question 3: List the details of all players who did not play any games.
SELECT T.*,P.Global_Level,P.Game_Played,P.User_Email,P.Username
FROM PlayerInfo P INNER JOIN Player2 T ON P.Player_Name = T.Player_Name
WHERE Game_Played =0
```

90 %

Results Messages

PlayerID	Player_Name	Online_Status	Battle_Points	Diamonds_Collected	Global_Level	Game_Played	User_Email	Username
----------	-------------	---------------	---------------	--------------------	--------------	-------------	------------	----------

Student 4: GOH WEI SONG (TP063479)

Question1: List the names and battle points of all players with the highest battle points. Arrange the list alphabetically according to the names.

Solution:

```
SELECT Player_Name, MAX(Battle_Points) AS 'Highest Battle Points'
FROM Player2
GROUP BY Player_Name
ORDER BY Player_Name ASC;
```

--Question1: List the names and battle points of all players with the highest battle points. Arrange the list alphabetically according to the names.

```
SELECT Player_Name, MAX(Battle_Points) AS 'Highest Battle Points'
FROM Player2
GROUP BY Player_Name
ORDER BY Player_Name ASC;
```

	Player_Name	Highest Battle Points
1	Abram Cole	19
2	Allison Baker	21
3	Ashley Young	16
4	Bobbi Pope	25
5	Chris Wade	44
6	Chung Garcia	19
7	Daniel Davis	25
8	Drew Ryan	5
9	Emily Chark	15
10	Eric Cantona	22
11	Eric White	11
12	Jack Lee	22
13	John Alex	39
14	Josh Pieters	33
15	Louis James	7
16	Raynold Matt	22
17	Reyna Tapia	7
18	Rivhard Blair	5
19	Waylon Long	12
20	Zane Gomez	21

Question 2: List the total number of games played in each mode on '15/3/2021'. Sort your answer in descending order of total number of games played.

Solution:

```
SELECT G.Game_Date, M.Mode_ID, COUNT(Game_ID) AS 'Total number of
games played'
FROM Game_Mode2 M INNER JOIN Game G ON M.Mode_ID = G.Mode_ID
WHERE Game_Date = '2021-03-15'
GROUP BY M.Mode_ID, G.Game_Date
ORDER BY COUNT(Game_ID) DESC;
```

```
--Question 2: List the total number of games played in each mode on '15/3/2021'. Sort your answer in descending order of total number of games played.
SELECT G.Game_Date,M.Mode_ID,COUNT(Game_ID) AS 'Total number of games played'
FROM Game_Mode2 M INNER JOIN Game G ON M.Mode_ID = G.Mode_ID
WHERE Game_Date = '2021-03-15'
GROUP BY M.Mode_ID,G.Game_Date
ORDER BY COUNT(Game_ID) DESC;
```

90 %

Results Messages

	Game_Date	Mode_ID	Total number of games played
1	2021-03-15	GM003	2
2	2021-03-15	GM001	1
3	2021-03-15	GM002	1

Question 3: List the id and name of the most used hero in all games.

Solution:

```
SELECT TOP 1 PH.Hero_ID,Hero_Name, COUNT(Player_ID) AS 'HeroUsed'
FROM Player_Hero PH JOIN Hero2 H ON H.Hero_ID=PH.Hero_ID
GROUP BY PH.Hero_ID,Hero_Name
ORDER BY HeroUsed DESC;
```

```
--Question 3: List the id and name of the most used hero in all games.
SELECT TOP 1 PH.Hero_ID,Hero_Name, COUNT(Player_ID) AS 'HeroUsed'
FROM Player_Hero PH JOIN Hero2 H ON H.Hero_ID=PH.Hero_ID
GROUP BY PH.Hero_ID,Hero_Name
ORDER BY HeroUsed DESC;
```

90 %

Results Messages

	Hero_ID	Hero_Name	HeroUsed
1	102	Hestia	3

11. Conclusion

To sum up, we created a database for an open mobile multiplayer online battle arena that includes a different type of heroes, effective pairing, and quicker combat that allows for rapid gameplay sessions on the go. It has many of the features that players would expect from a multi-player battle arena game, such as hero roles, abilities, skins, and more. The game allows for a flawless gameplay experience on tablets and phones, with some adjustable features such as auto-aiming, last-hitting, and more to help control the game. (Moonton, 2016)

References:

Moonton. (2016, July 14). *SandBox MMORPG*. Retrieved from Mobile Legends:
<https://mmos.com/review/mobile-legends>

W3 Schools. (n.d.). SQL SELECT TOP, LIMIT, ROWNUM.
https://www.w3schools.com/sql/sql_ref_top.asp

Workload Matrix

No.	Task Components	Abdullah Al Mukit	Saw Chiou Ming	S M Rabeya Amin Ikra	Goh Wei Song
1.	Introduction	25%	25%	25%	25%
2.	Final ERD	25%	25%	25%	25%
3.	Mapping of ERD	25%	25%	25%	25%
4.	Normalization	25%	25%	25%	25%
5.	Relational Schema in 3NF	25%	25%	25%	25%
6.	Data Dictionary	25%	25%	25%	25%
7.	Database Diagram	25%	25%	25%	25%
8.	DDL Statements	25%	25%	25%	25%
9.	Insert Statements	25%	25%	25%	25%
10.	DML	25%	25%	25%	25%
	Student 1	100%			
	Student 2		100%		
	Student 3			100%	
	Student 4				100%
11.	Conclusion	25%	25%	25%	25%
	Signature	Mukit		<i>Ikra</i>	<i>Song</i>